CyberSight AI

Intelligent Firewall Log Analysis Report

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CYBERSECURITY FIREWALL TRAFFIC & THREAT ANALYSIS REPORT

1. Threat Level

HIGH

2. Risk Score

75 / 100

3. Executive Summary

This report presents a security analysis of recent firewall telemetry comprising 500 logged connection events and approximately 37.88 MB ($37,877,294$ bytes) of total network traffic.

The analysis reveals an elevated risk profile characterized by a high firewall block rate (38.2%), significant targeting of high-risk administrative and file-sharing services (SMB, RDP, SSH), and abnormal ICMP volume indicative of network reconnaissance. While active controls are successfully filtering unauthorized traffic, the heavy concentration of inbound/outbound connection requests on critical management ports requires immediate mitigation to prevent potential perimeter breach, brute-force compromises, or ransomware deployment.

4. Key Findings

* High Block Rate (38.2%): Out of 500 connection attempts, 191 were blocked while 309 were allowed. This high rejection ratio signals active scanning, policy violations, or automated malicious probes reaching the network perimeter.

* Targeting of Critical & High-Risk Ports:

* Port 445 (SMB - 59 requests): Top targeted destination. SMB exposure poses extreme risks related to lateral movement, unauthorized file share access, and ransomware propagation.

* Port 3389 (RDP - 58 requests) & Port 22 (SSH - 55 requests): Combined total of 113 connection attempts targeting remote management protocols, indicating intense interest in gaining administrative access.

* Abnormal ICMP Protocol Volume: ICMP accounts for 31.8% (159 connection events) of total protocol distribution (TCP: 174, UDP: 167, ICMP: 159). This disproportionately high volume strongly suggests host discovery sweeps, ping sweeps, or potential covert channel activity.

* Cleartext Protocol Exposure: Significant activity observed on unencrypted legacy protocols, including IMAP (Port 143 - 47 events), HTTP (Port 80 - 48 events), and POP3 (Port 110 - 39 events), exposing credentials and sensitive data to potential interception.

* Geographic Spread: Traffic originates across 7 distinct countries, heavily led by the UAE (84), Japan (80), Germany (79), UK (71), and the USA (67).

5. Risk Assessment

| Risk Domain | Identified Threat | Severity | Impact |

| :--- | :--- | :--- | :--- |

| Ransomware & Exploitation | Excessive activity on Port 445 (SMB) and Port 3389 (RDP). | CRITICAL | High probability of automated exploit attempts (e.g., EternalBlue, BlueKeep variants) or brute-force access resulting in domain-wide compromise. |

| Network Reconnaissance | High ICMP traffic (159 events) and broad port probing. | MEDIUM-HIGH | Threat actors actively mapping internal network topology and identifying listening services. |

| Credential Harvesting | Use of unencrypted protocols (POP3/110, IMAP/143, HTTP/80). | MEDIUM | Eavesdropping or Man-in-the-Middle (MitM) attacks leading to cleartext credential theft. |

| Unauthorized Access | Exposure/probe of Port 22 (SSH) and non-standard web ports (8080). | MEDIUM-HIGH | Exposure of management interfaces or unpatched web applications to external brute-force vectors. |

6. Security Recommendations

Immediate Action Items (0 - 24 Hours)

1. Restrict SMB & RDP Access: Ensure Ports 445 and 3389 are strictly prohibited from exposure to the public internet. Restrict RDP and SSH access exclusively through a hardened Virtual Private Network (VPN) or Zero Trust Network Access (ZTNA) solution requiring Multi-Factor Authentication (MFA).

2. Implement ICMP Rate-Limiting: Configure firewall rules to drop or rate-limit inbound and outbound ICMP traffic (pings/echo requests) to neutralize active reconnaissance sweeps.

3. Audit Blocked & Allowed Logs for Port 445/3389: Verify whether the 309 *ALLOWED* connections included traffic on Ports 445, 3389, or 22. Immediately isolate any external IP addresses attempting direct communication with these ports.

Strategic Improvements (1 - 2 Weeks)

1. Disable Legacy Mail Protocols: Enforce mandatory TLS encryption by migrating all email clients from cleartext IMAP (143) and POP3 (110) to IMAPS (993) and POP3S (995).

2. Geo-IP Filtering: Enforce strict ingress Geo-IP blocking for geographic regions where the organization maintains no legitimate operational or business presence.

3. Deploy Intrusion Prevention Systems (IPS): Enable automated IPS signatures to detect and dynamically drop brute-force attacks against SSH (22), RDP (3389), and HTTP/S management interfaces.

7. Overall Security Status

DEFENSIVE POSTURE: DEGRADED / UNDER ACTIVE PROBING

The security boundary is actively stopping a high volume of unauthorized attempts (38.2% block rate). However, the network's current attack surface exhibits unacceptable exposure across critical management and remote access protocols (SMB, RDP, SSH).

While the firewall is functioning as designed, the high volume of targeted probing indicates that external actors are actively hunting for vulnerabilities within the network infrastructure. Implementation of the outlined recommendations—specifically perimeter hardening of SMB/RDP and disabling unencrypted protocols—is required immediately to shift the overall risk status back to an acceptable baseline.